- $\Theta^* = \arg\min_{\Theta} J(\Theta)$
- $\Theta^{(t)} = \Theta^{(t-1)} - \eta f'(\Theta^{(t-1)})$
- $\nabla_\Theta J = \begin{bmatrix} \partial J / \partial \Theta_1 \\ \ddots \\ \partial J / \partial \Theta_m \end{bmatrix}$ $\quad \Theta^{(t)} = \Theta^{(t-1)} - \eta \nabla_\Theta J(\Theta^{(t-1)})$

Note. 종료 조건

- $t = T$
- $|\Theta^{(t)} - \Theta^{(t-1)}| < \varepsilon$
- $|f'(\Theta^{(t)})| < \varepsilon$
- $|J(\Theta^{(t)}) - J(\Theta^{(t-1)})| < \varepsilon$

$$g^{(i)} = \sigma(\theta^T x^{(i)} + \theta_0)$$

$$\mathcal{L}_{nll}(g, y) = -(y \log g + (1-y) \log(1-g))$$

$$J_{lr}(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}_{nll}(g^{(i)}, y^{(i)}) + \frac{\lambda}{2} \|\theta\|^2$$

$$\nabla_\theta J_{lr} = \frac{1}{n} \sum_{i=1}^{n} (g^{(i)} - y^{(i)}) x^{(i)} + \lambda \theta$$

$$\frac{\partial J_{lr}}{\partial \theta_0} = \frac{1}{n} \sum_{i=1}^{n} (g^{(i)} - y^{(i)})$$

Note.

$$\nabla_\theta g = g(1-g) \cdot x^{(i)}$$

$$\frac{\partial \mathcal{L}}{\partial g} = -\frac{y}{g} + \frac{1-y}{1-g}$$

$$\nabla_\theta \mathcal{L} = \left(y(g-1) + g(1-y)\right) \cdot x^{(i)}$$

$$= (g-y) \cdot x^{(i)}$$

- GD : $f(\Theta) = \sum_i f_i(\Theta)$
- SGD : $\Theta^{(t+1)} = \Theta^{(t)} - \eta_t \nabla_t f_{n_t}(\Theta^{(t)})$

$$L_n = \frac{1}{2}(y^{(n)} - \hat{y}^{(n)})^2$$

$$w_i^{(t+1)} = w_i^{(t)} - \eta \frac{\partial L_n}{\partial w_i}$$

- Mini-Batch : $L_B = \frac{1}{2} \sum_{n \in B} (y^{(n)} - \hat{y}^{(n)})^2$
- SGD 수렴조건 : $\sum_{t=1}^{\infty} \eta(t) = \infty, \quad \sum_{t=1}^{\infty} \eta(t)^2 < \infty$

Momentum

- $v_i^{(t)} = \alpha v_i^{(t)} - \eta \frac{\partial L}{\partial w_i^{(t)}}$
- $w^{(t+1)} = w^{(t)} + v^{(t)}$

Ada Grad

- $h \leftarrow h + \frac{\partial L}{\partial W} \odot \frac{\partial L}{\partial W}$
- $W \leftarrow W - \frac{\eta}{\sqrt{h+\varepsilon}} \frac{\partial L}{\partial W}$

Adam

- $g_t = \nabla_\theta f_t(\theta_{t-1})$
- $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$
- $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$
- $\hat{m}_t = \dfrac{m_t}{1-\beta_1^t}$, $\hat{v}_t = \dfrac{v_t}{1-\beta_2^t}$
- $\theta_t = \theta_{t-1} - \alpha \dfrac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon}$

Back-Propagation

- $J = M_k(M_{k-1}(\cdots M_1(x)))$
- $u^{[0]} = x,\quad u^{[i]} = M_i(u^{[i-1]}),\quad u^{[k]} = J$
- $\dfrac{\partial J}{\partial z_i} = \displaystyle\sum_{j=1}^{n} \underbrace{\frac{\partial J}{\partial u_j}}_{\text{upstream}} \cdot \frac{\partial u_j}{\partial z_i} = B[g, z](v)$

$$\begin{cases} u = g(z) \\ v = \partial J / \partial u \end{cases}$$

Linear Layer

$u = Wz + b$

$B[MM, z](v) = W^T v$

$B[MM, W](v) = v z^T$

$B[MM, b](v) = v$

Activation

$B[\sigma, z](v) = \sigma'(z) \odot v$

ReLU

$\mathrm{ReLU}(z) = \max(0, z)$

$$\mathrm{ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0 & z < 0 \end{cases}$$

Squared Loss

$l_{MSE} = \frac{1}{2}(z-y)^2$

$B[l_{MSE}, z](v) = (z-y)v$

Logistic Loss

$$l_{logistic}(t, y) = y \log(1+\exp(-t)) + (1-y)\log(1+\exp(t))$$

$$B[l_{logistic}, t](v) = \left(\frac{1}{1+\exp(-t)} - y\right) v$$

Forward $z^{[1]} = MM_{w^{[1]}, b^{[1]}}(x)$. $a^{[1]} = \sigma(z^{[1]})$

$z^{[2]} = MM_{w^{[2]}, b^{[2]}}(a^{[1]})$, $a^{[2]} = \sigma(z^{[2]})$

$\cdots$

$z^{[r]} = MM_{w^{[r]} b^{[r]}}(a^{[r-1]})$, $J = \ell_{logistic}(z^{[r]}, y)$

Backward

$$\frac{\partial J}{\partial z^{[r]}} = \frac{1}{1+\exp(-z^{[r]})} - y \qquad \frac{\partial J}{\partial W^{[k]}} = \frac{\partial J}{\partial z^{[k]}} \cdot (a^{[l-1]})^T$$

$$\frac{\partial J}{\partial a^{[k]}} = (W^{[k+1]})^T \frac{\partial J}{\partial z^{[k+1]}} \qquad \frac{\partial J}{\partial b^{[k]}} = \frac{\partial J}{\partial z^{[k]}}$$

$$\frac{\partial J}{\partial z^{[k]}} = \sigma'(z^{[k]}) \odot \frac{\partial J}{\partial a^{[k]}}$$

- $S_n = \{(\underset{\mathbb{R}^n}{\underset{\downarrow}{x^{(1)}}}, \underset{\mathbb{R}}{\underset{\downarrow}{y^{(1)}}}), \ldots\ldots, (x^{(n)}, y^{(n)})\}$

- Loss $= (\text{guess} - \text{actual})^2$

- 선형회귀 가설 $h(x) = \theta^T \phi(x) + \theta_0$

  $J(\theta, \theta_0) = \frac{1}{n} \sum_i (\theta^T x^{(i)} + \theta_0 - y^{(i)})^2$

- $x \leftarrow \begin{bmatrix} x \\ 1 \end{bmatrix}$ then $h(x) = \theta^T x$

- $W = \begin{bmatrix} x^{(1)T} \\ \ddots \\ x^{(n)T} \end{bmatrix} \in \mathbb{R}^{n \times d}$ $T = \begin{bmatrix} y^{(1)} \\ \ddots \\ y^{(n)} \end{bmatrix} \in \mathbb{R}^{n \times 1}$

- 예측 $W\theta$

  2정규화

  목적함수 $J = \frac{1}{n}(W\theta - T)^T (W\theta - T) \longrightarrow \frac{1}{n}(W\theta - T)^T (W\theta - T) + \lambda \|\theta\|^2$

  $\nabla_\theta J = \frac{2}{n} W^T (W\theta - T) = 0$ 최소값. $\longrightarrow \frac{2}{n} W^T (W\theta - T) + 2\lambda\theta = 0$

  $\underbrace{W^T W}\theta_{OLS} = W^T T$ $(W^T W + n\lambda I)\theta = W^T T$

  역행렬이 없다면? 과적합?

  역행렬 $O(d^3)$

  $\rightsquigarrow$ 경사하강법

Test Error = Structural Error + Estimation Error

가설공간 크기가 부족하다. 데이터가 부족하다.

- $\lambda \Uparrow$: structural $\Uparrow$ estimation $\Downarrow$
- $\lambda \Downarrow$: structural $\Downarrow$ estimation $\Uparrow$

다항식 특성변환

- $\phi_k(x) = [1, x, x^2, \ldots, x^k]$: $k \Uparrow$ then structural $\Downarrow$ estimation $\Uparrow$

K-means

- $c^{(i)} := \arg\min_j \|x^{(i)} - \mu_j\|^2$
- $\mu_j := \dfrac{\sum_{i=1}^{n} 1\{c^{(i)} = j\}\, x^{(i)}}{\sum_{i=1}^{n} 1\{c^{(i)} = j\}}$
- $J(c, \mu) = \sum_{i=1}^{n} \|x^{(i)} - \mu_{c^{(i)}}\|^2$

- J를 $c \to \mu \to c \cdots$로 번갈아 줄인다.
- 단조감소. $\longrightarrow \boxed{\text{local min}}$

여러 초기값.

Gaussian Mixture Model

- $z^{(i)} \sim \text{Multinomial}(\phi)$ with $\phi_j = p(z^{(i)} = j)$
- $x^{(i)} \mid z^{(i)} = j \sim N(\mu_j, \Sigma_j)$
- $p(x^{(i)}, z^{(i)}) = p(x^{(i)} \mid z^{(i)})\, p(z^{(i)})$
- $\ell(\phi, \mu, \Sigma) = \sum_{i=1}^{n} \underbrace{\log \sum_{j=1}^{n} \phi_j N(x^{(i)}; \mu_j, \Sigma_j)}_{\text{어려움.}}$

$z^{(i)}$를 안다면

$$\phi_j = \frac{1}{n} \sum_i 1\{z^{(i)} = j\}$$

$$\mu_j = \frac{\sum_i 1\{z^{(i)} = j\}\, x^{(i)}}{\sum_i 1\{z^{(i)} = j\}}$$

$$\Sigma_j = \frac{\sum_i 1\{z^{(i)} = j\}(x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^T}{\sum_i 1\{z^{(i)} = j\}}$$

EM은 확률 $w_j^{(i)}$를 이용한다.

$$w_j^{(i)} = p(z^{(i)} = j \mid x^{(i)}) = \frac{\phi_j N(x^{(i)}: \mu_j, \Sigma_j)}{\sum_l \phi_l N(x^{(i)}: \mu_l, \Sigma_l)}$$

$$\phi_j = \frac{1}{n} \sum_i w_j^{(i)}$$

$$\mu_j = \frac{\sum_i w_j^{(i)} x^{(i)}}{\sum_i w_j^{(i)}}$$

$$\Sigma_j = \frac{\sum_i w_j^{(i)} (x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^T}{\sum_i w_j^{(i)}}$$

일반 EM

- $$\begin{aligned} \log p(x^{(i)}; \theta) &= \log \sum_z p(x^{(i)}, z; \theta) \\ &= \log \sum_z Q(z) \cdot \frac{p(x^{(i)}, z; \theta)}{Q(z)} \\ &\boxed{\geq} \sum_z Q(z) \log \frac{p(x^{(i)}, z; \theta)}{Q(z)} = \mathrm{ELBO}(x; Q, \theta) \end{aligned}$$

등호 iff $Q(z) = p(z \mid x; \theta)$

- $$\begin{aligned} \ell(\theta) &= \sum_i \log p(x^{(i)}; \theta) \\ &\geq \sum_i \mathrm{ELBO}(x^{(i)}; Q, \theta) \end{aligned}$$

- E-Step : $Q_i^{(t)}(z^{(i)}) = p(z^{(i)} \mid x^{(i)}; \theta^{(t)})$

- M-Step : $\theta^{(t+1)} = \underset{\theta}{\arg\max} \sum_i \mathrm{ELBO}(x^{(i)}; Q_i^{(t)}, \theta)$

- $$\begin{aligned} \ell(\theta^{(t+1)}) &\geq \sum_i \mathrm{ELBO}(x^{(i)}; Q_i^{(t)}, \theta^{(t+1)}) \\ &\geq \sum_i \mathrm{ELBO}(x^{(i)}; Q_i^{(t)}, \theta^{(t)}) = \ell(\theta^{(t)}) \end{aligned}$$

likelyhood가
증가한다.

$$= \log p(x; \theta) - \boxed{D_{KL}(Q \parallel p_{z|x})}$$

작아야 함.

$$\begin{aligned} \mathrm{ELBO}(x; Q, \theta) &= \mathbb{E}_Q[\log p(x, z; \theta)] - \mathbb{E}_Q[\log Q(z)] \\ &= \mathbb{E}_Q[\log p(x \mid z; \theta)] - D_{KL}(Q \parallel p_z) \end{aligned}$$

VAE

- $z \sim \mathcal{N}(0, I_k)$
- $x \mid z \sim \mathcal{N}(g(z;\theta), \sigma^2 I_d)$
- $p(z \mid x;\theta) \underset{(\text{근사})}{\approx} Q_i = \mathcal{N}(g_\phi(x^{(i)}), \mathrm{diag}(v_\psi(x^{(i)}))^2)$
- $\mathrm{ELBO}(\phi, \psi, \theta) = \sum_{i=1}^{n} \mathbb{E}_{z^i \sim Q_i}\left[\log \frac{p(x^{(i)}, z^{(i)};\theta)}{Q_i(z^{(i)})}\right]$
- 업데이트

$$\nabla \mathbb{E}_{z \sim Q_\phi}[f(\phi)] \neq \mathbb{E}_{z \sim Q_\phi}[\nabla f(\phi)]$$

$$\theta := \theta + \eta \nabla_\theta \mathrm{ELBO}$$

$$\phi := \phi + \eta \nabla_\phi \mathrm{ELBO}$$

$$\psi := \psi + \eta \nabla_\psi \mathrm{ELBO}$$

$$z^{(i)} = g(x^{(i)};\phi) + v(x^{(i)};\psi) \odot \underbrace{\varepsilon^{(i)}}_{\sim \mathcal{N}(0, I_k)}$$

$\Rightarrow \mathbb{E}_{z^{(i)} \sim Q_i}[\cdots]$를 $\mathbb{E}_{\varepsilon^{(i)} \sim \mathcal{N}(0, I)}[\cdots]$로 변경.

전처리 $x_j^{(i)} \leftarrow \dfrac{x_j^{(i)} - \mu_j}{\sigma_j}$

$$\mu_j^{(i)} \longleftarrow \frac{1}{n}\sum_{i=1}^{n} x_j^{(i)}$$

$$\sigma_j^{(i)} \longleftarrow \frac{1}{n}\sum_{i=1}^{n} (x_j^{(i)} - \mu_j)^2$$

PCA는 단위벡터 u에 대한 투영분산을 최대화.

- $\max_{\|u\|=1} \frac{1}{n}\sum_{i=1}^{n} ((x^{(i)})^T u)^2 = \max_{\|u\|=1} u^T \Big(\underbrace{\frac{1}{n}\sum_{i=1}^{n} x^{(i)} x^{(i)T}}_{=: \Sigma}\Big) u$

k차원에서는

top-k eigenvectors

- $L(u, \lambda) = u^T \Sigma u - \lambda(u^T u - 1)$
- $\nabla_u L = 2\Sigma u - 2\lambda u = 0$
- $\Sigma e = \lambda e$

$\Rightarrow u = \sum_i a_i \overset{\text{고유벡터}}{e_i}$

가장 큰 고유값 선택.

$$u^T \Sigma u = \sum_i \lambda_i^2 a_i^2$$

state machine.

- $\mu = (S, \underset{\text{input}}{\textcircled{X}}, \underset{\text{output}}{\textcircled{Y}}, s_0, f, g)$

- $s_t = f(s_{t-1}, x_t)$

- $y_t = g(s_t)$

- LTI (선형) $y_t = 3y_{t-1} + 6y_{t-2}$
  $+ 5x_t + 3x_{t-2}$

- RNN (비선형) $s_t = f_1(W^{sx}x_t + W^{ss}s_{t-1} + W_0^{ss})$
  $y_t = f_2(W^{o}s_t + W_0^{o})$

- $MDP = \langle S, A, T, R, \gamma \rangle$

  ① $T(s, a, s') = P(S_t = s' \mid S_{t-1} = s, A_{t-1} = a)$

  ② $R : S \times A \rightarrow \mathbb{R}$

  ③ $r \in [0, 1]$

  ④ $\pi : S \rightarrow A$

- $V_\pi^0(s) = 0$

- $V_\pi^k(s) = R(s, \pi(s)) + \sum_{s'} T(s, \pi(s), s') V_\pi^{k-1}(s')$

- $\pi_1 \overset{h}{>} \pi_2$ means ① $\forall s \quad V_{\pi_1}^k(s) \geq V_{\pi_2}^h(s)$

  ② $\exists s \quad V_{\pi_1}^h(s) > V_{\pi_2}^h(s)$

---

finite horizon

- $Q^0(s, a) = 0$

- $Q^h(s, a) = R(s, a) + \sum_{s'} T(s, a, s') \max_{a'} Q^{h-1}(s', a')$

- $\pi_h^*(s) = \arg\max_a Q^h(s, a)$

infinite horizon

- $0 < \gamma < 1$
- $J_\pi(s_0) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t \ : \ \pi, \ S_0 = s_0 \right]$
- 기대 생존 길이 $\frac{1}{1-\gamma}$

Bellman Equation

- $V_\pi(s) = \mathbb{E}\left[R_0 + \gamma R_1 + \gamma^2 R_2 + \cdots \mid \pi, S_0 = s \right]$ ⇒ 선형방정식.

  $= R(s, \pi(s)) + \gamma \sum_{s'} T(s, \pi(s), s') V_\pi(s')$

- $\exists \pi^*$ s.t $\forall s$ $\forall \pi$ $V_{\pi^*}(s) \geq V_\pi(s)$
- $Q^*(s,a) = R(s,a) + \gamma \sum_{s} T(s,a,s') \max_{a'} Q^*(s',a')$
- $\pi^*(s) = \arg\max_{a} Q^*(s,a)$

- 초기화 $Q_{old}(s,a) = 0$
- 반복 업데이트 $Q_{new}(s,a) = R(s,a) + \gamma \sum_{s'} T(s,a,s') \max_{a'} Q_{old}(s',a')$
- 종료 조건 $\max_{s,a} \left| Q_{old}(s,a) - Q_{new}(s,a) \right| < \varepsilon$
- $\pi_Q(s) = \arg\max_{a} Q(s,a)$
- $\| V_{\pi_Q} - V_{\pi^*} \|_{max} < \varepsilon$

Note. 충분히 작은 ε에 대해

$$\| Q_{old} - Q_{new} \|_{max} < \varepsilon \Rightarrow \pi_Q = \pi^*$$

RL/MDP의 기본식

- $M = (S, A, \{P_{sa}\}, \gamma, R)$
- $s_{t+1} \sim P_{s_t a_t}$, $\pi : S \to A$
- $J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$

- $V^\pi(s) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right]$

  $= R(s) + \gamma \sum_{s' \in S} P_{s,\pi(s)}(s') V^\pi(s')$

- $V^*(s) = \max_\pi V^\pi(s)$
- $V^*(s) = \max_a \left[R(s,a) + \gamma \mathbb{E}_{s' \sim P_{sa}} V^*(s') \right]$
- $\pi^*(s) = \arg\max_a \left[R(s,a) + \gamma \mathbb{E}_{s' \sim P_{sa}} V^*(s') \right]$

Value Iteration

- $B[V](s) = \max_a \left[R(s,a) + \gamma \mathbb{E}_{s' \sim P_{sa}} [V(s')] \right]$
- $V_{k+1} = B[V_k] \longrightarrow V^*$
- $\| V_{k+1} - V^* \|_\infty = \| B(V_k) - V^* \|_\infty \leq \gamma \| V_k - V^* \|_\infty$

Policy Iteration

- $V^\pi = R^\pi + \gamma P^\pi V^\pi$ → 반복횟수 $\begin{cases} = 1 & \text{value iteration} \\ = \infty & \text{policy iteration} \end{cases}$
- $\pi(s) \longleftarrow \arg\max_a \sum_{s'} P_{sa}(s') \cdot V^\pi(s')$

Finite-Horizon MDP

- $M = (S, A, P_{sa}^{(t)}, T, R^{(t)})$
- $J = \mathbb{E}\left[\sum_{t=0}^{T} R^{(t)}(s_t, a_t) \right]$
- $\pi^{(t)} : S \rightarrow A$
- $V_t^*(s) = \max_a \left[R^{(t)}(s,a) + \mathbb{E}_{s' \sim P_{sa}^{(t)}} \left[V_{t+1}^*(s') \right] \right]$

LQR

- $s_{t+1} = A_t s_t + B_t a_t + \boxed{w_t}$ $\sim \mathcal{N}(0, \Sigma_t)$
- $R^{(t)}(s_t, a_t) = -s_t^T U_t s_t - a_t^T W_t a_t$
- $V_t^*(s_t) = s_t^T \Phi_t s_t + \Psi_t$
- $a_t^* = L_t s_t$

Note.

- Σ_t는 Ψ_t에만 영향.
- L_t는 A_t, B_t, Φ_{t+1}로 결정